

Sustainable Software Engineering



Measuring Socio-Technical Aspects in Software Development Teams



GUIDO

Gathering and Understanding Socio-Technical Aspects
in Development Organizations

What are Community Smells?

"Socio-technical anti-patterns that act as early indicators of issues potentially leading to increased project costs or failure."



These are among the most-known social debt instances, and they focus on issues related to *communication and collaboration* within software development teams

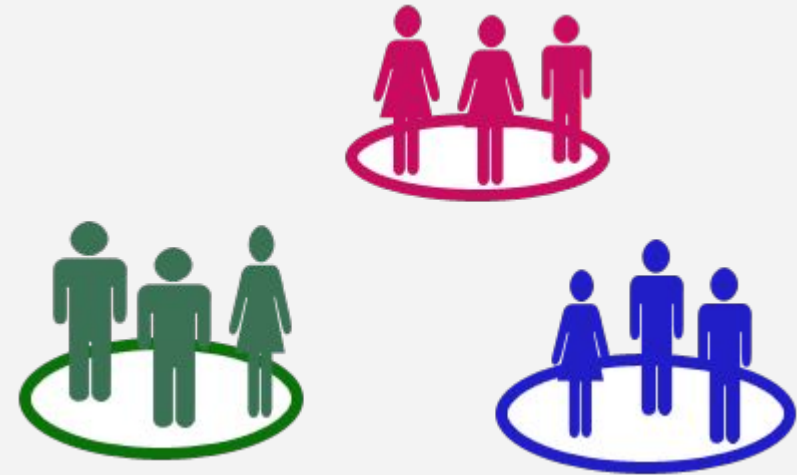
Smell #1: Organizational Silo

Definition

Siloed areas of the community that do not communicate, except through one or two of their respective members.

Impact

- Bottlenecks in information flow.
- "Us vs. Them" mentality.
- Duplication of effort.
- High "Truck Factor" risk (if the bridge leaves, the connection breaks).



Smell #2: Black Cloud



Definition

Information overload due to a lack of structured communications or cooperation governance.

Impact

- Team members are bombarded with irrelevant data.
- Critical information gets lost in the noise.
- Decision paralysis and cognitive fatigue.
- Reduced productivity due to constant context switching.

Smell #3: Prima Donnas

Definition

A team or individual unwilling to respect external changes or feedback from other members due to an inefficiently structured collaboration.

Impact

- Resistance to code reviews or architectural changes.
- Toxic environment for new contributors.
- Technical stagnation due to refusal to adapt.



Smell #4: Radio Silence

The Pattern

One member interposes themselves into every formal interaction across sub-communities.

There is little flexibility to introduce other communication channels.

The Consequence

This creates a rigid communication structure dependent on a single node. It stifles organic collaboration and slows down decision-making processes, as all info must pass through the "gatekeeper."

Complete Community Smell Catalog

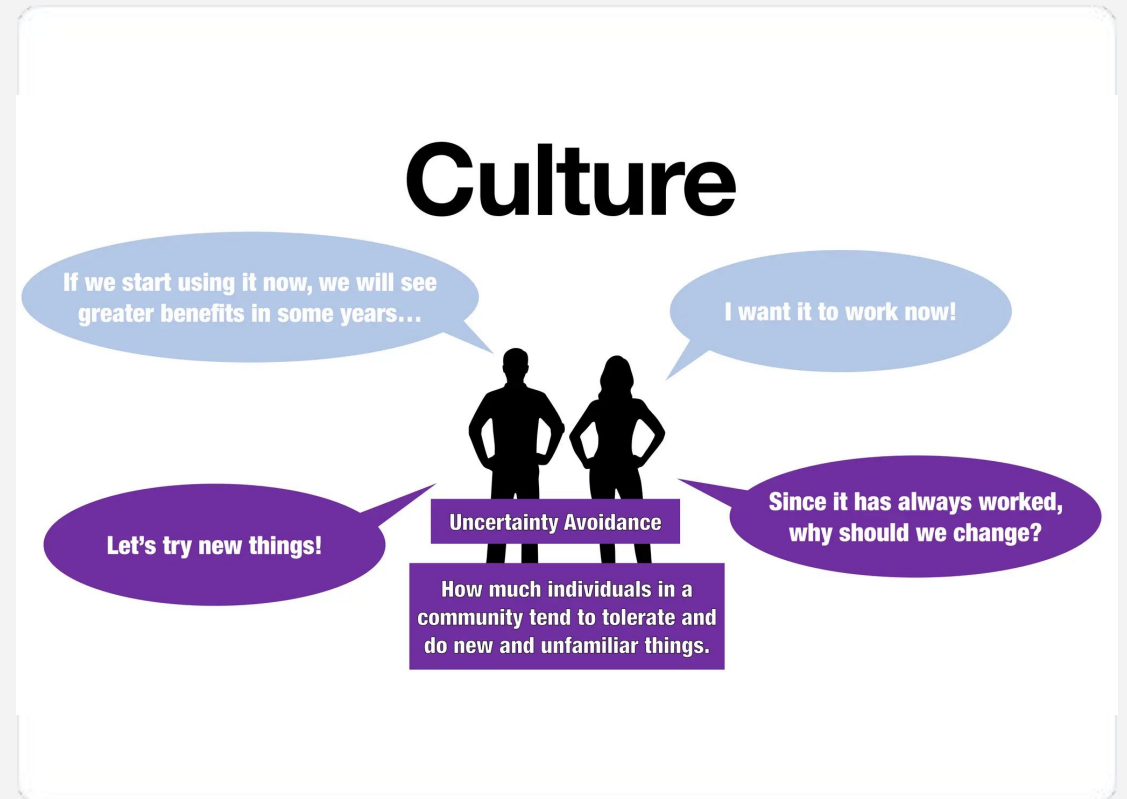
Smell Name	Description
Sharing Villainy	Team members share essential knowledge that is outdated, wrong, or unconfirmed due to lack of exchange.
Organizational Skirmish	Misalignment between expertise levels leads to dropped productivity and increased costs.
Solution Defiance	Division into conflicting subgroups due to different cultural/experience backgrounds.
Truck Factor Smell	Risk of significant knowledge loss if key developers (who hold all the info) leave the project.
Unhealthy Interaction	Long delays in stakeholder communications cause slow, light, and brief discussions.
Toxic Communication	Toxic interactions and conflicting opinions push developers to leave the project.

Cultural Dispersion

Another indicator of poor collaboration in teams is given by **cultural dispersion**. It indicates cultural differences within development teams, and it is based on frameworks such as **Hofstede's Cultural Dimensions**.

This framework scores cultures on dimensions like:

- Power Distance
- Individualism vs. Collectivism
- Uncertainty Avoidance
- Long-Term Orientation



Cultural Dispersion



High cultural dispersion is empirically correlated with the emergence of specific community smells.

Example: A clash in "Power Distance" between team members can lead to Organizational Silos or Radio Silence.

How Do We Detect Smells?



Mining Repositories

Tools mine commit history, issue trackers, and communication logs (co-commits, pull requests, etc.) to build a "Developer Social Network".



Social Network Analysis

Algorithms analyze the graph structure (nodes=devs, edges=interaction) to find patterns like disconnected clusters (Silos).



Machine Learning

Techniques like Genetic Algorithms and Decision Trees classify these patterns based on training datasets (e.g., CodeFace4Smells, csDetector).

Why New Tools Are Needed

csDetector

Pros: Strong detection capabilities.

Cons: Designed for researchers, command-line based, low usability for practitioners.

CADOCS

Pros: Conversational agent interface.

Cons: Tightly integrated with Slack (reduced flexibility), no cultural analysis features.

Neither tool combined smell detection with cultural insights.



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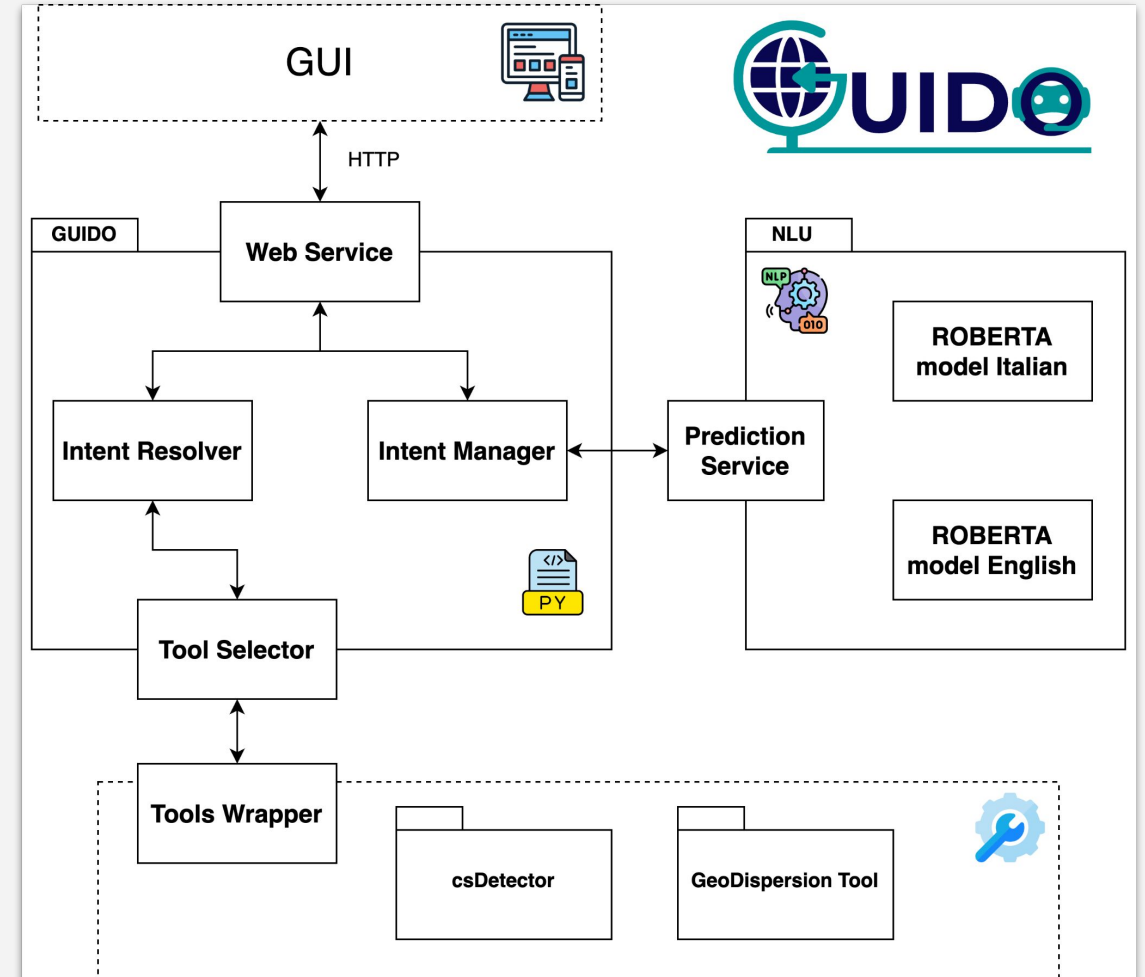
GUI DO Architecture

Modular Design: Client-server model for extensibility.

Intent Manager: Uses Natural Language Understanding (NLU) to parse user requests.

Strategy Pattern: Dynamically selects the right tool (csDetector or GeoDispersion) based on the intent.

Tool Wrapper: Encapsulates external analysis tools for unified access.



Core Functionalities



Detection (I1, I2)

Detect 10 community smells in GitHub repos (whole history or specific date range).



Mitigation (I3)

Suggest tailored refactoring strategies for identified smells to improve health.



Culture (I4)

Assess cultural dispersion and identify related socio-technical risks.

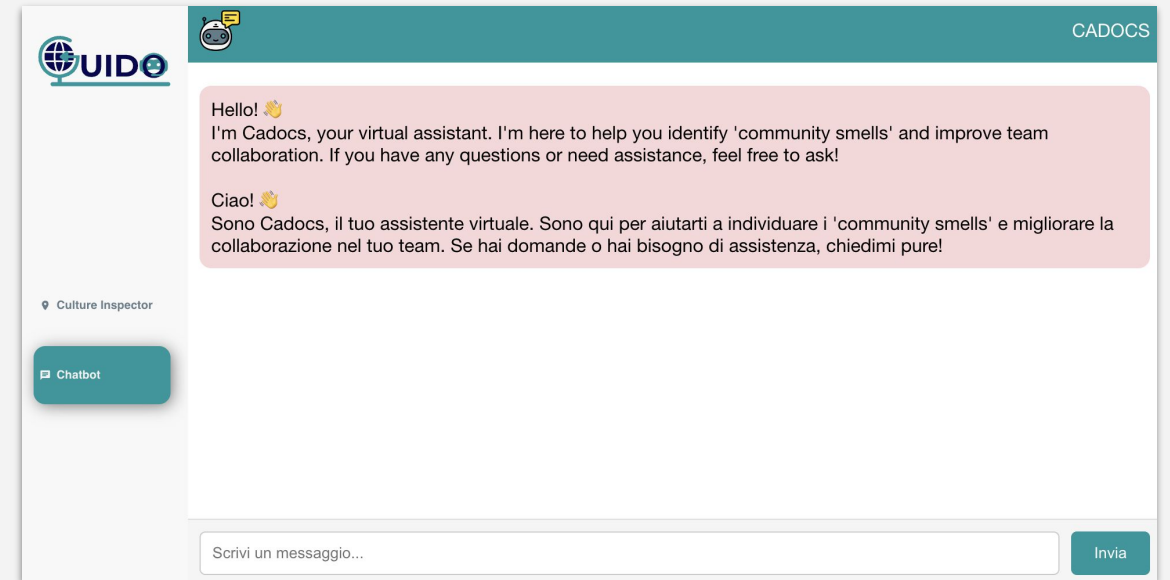
Conversational Interface

Natural Language Interaction

Users can simply type:

"Detect community smells in the repository [https://github.com/...](https://github.com/)"

The system uses NLU to understand the intent and triggers the analysis pipeline in the background.



Actionable Mitigation



Moving Beyond Detection

Detection is useless without a solution. GUiDO provides specific recommendations based on literature.

Example for "Organizational Silo":

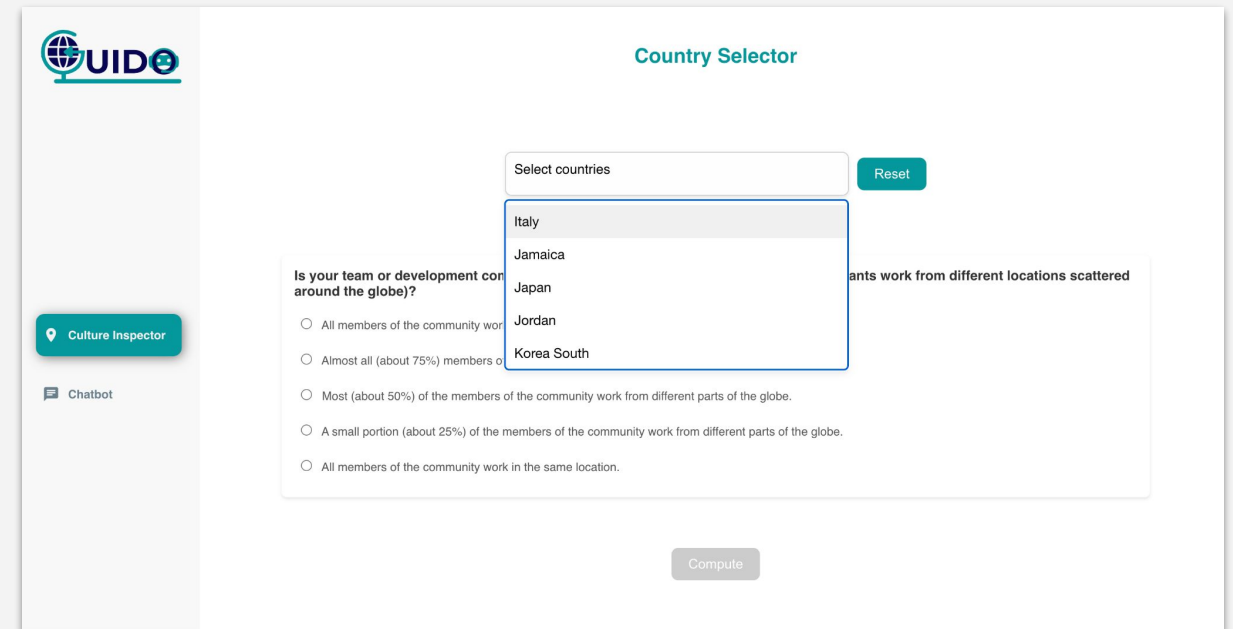
- "Introduce a rotation program for developers."
- "Organize cross-team hackathons."
- "Refactor communication channels to be public by default."

Cultural Analytics Dashboard

Visualizing Diversity

The dashboard allows managers to input team member locations.

It calculates dispersion metrics based on Hofstede's model and warns about potential friction points (e.g., "High risk of 'Radio Silence' due to Power Distance gap").



The screenshot displays the GUIDO Cultural Analytics Dashboard. On the left sidebar, there is a 'Culture Inspector' button with a location pin icon and a 'Chatbot' button with a speech bubble icon. The main content area is titled 'Country Selector' and features a 'Select countries' input field with a 'Reset' button to its right. A dropdown menu is open below the input field, listing 'Italy', 'Jamaica', 'Japan', 'Jordan', and 'Korea South'. Below the dropdown, a question is posed: 'Is your team or development community spread around the globe?'. Five radio button options are provided: 'All members of the community work from the same location.', 'Almost all (about 75%) members of the community work from different locations scattered around the globe.', 'Most (about 50%) of the members of the community work from different parts of the globe.', 'A small portion (about 25%) of the members of the community work from different parts of the globe.', and 'All members of the community work in the same location.' A 'Compute' button is located at the bottom right of the form.

Workflow 1: Diagnosing a Repo

1

Input

User pastes GitHub URL into Chat.

2

Analysis

GUI DO mines repo & builds social graph.

3

Report

Bot lists detected smells & severity.

4

Action

User asks "How do I fix this?"

Workflow 2: Team Assessment

1

Setup

Switch to Dashboard tab.

2

Data Entry

Select home countries of
developers.

3

Calculation

System computes Hofstede
distance.

4

Insight

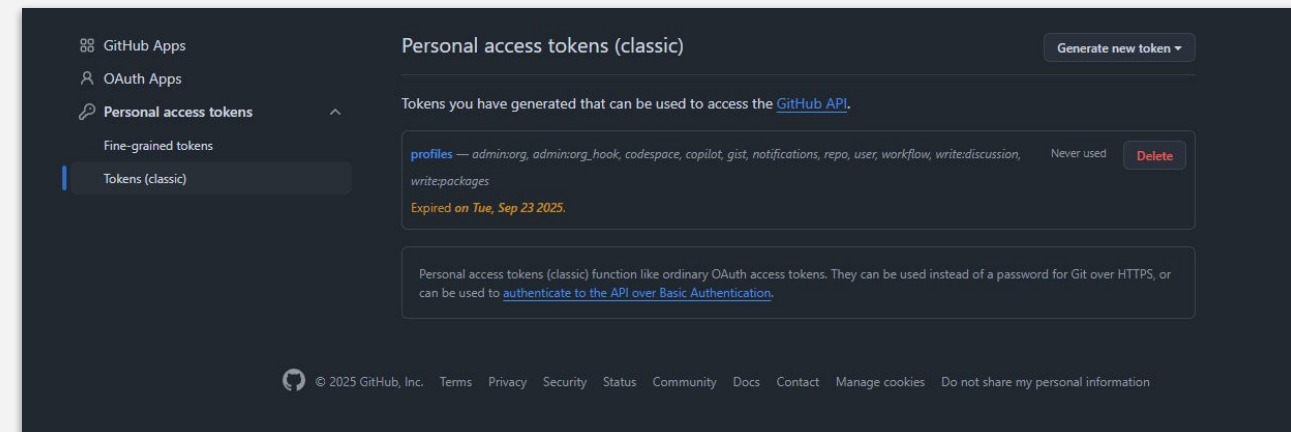
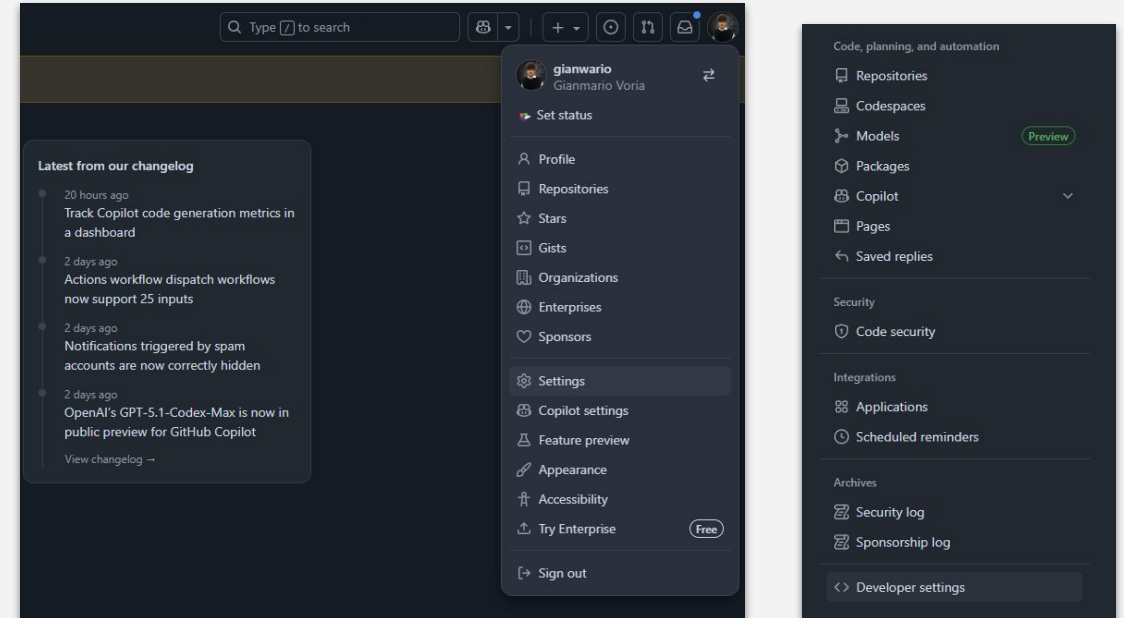
View risk map of potential smells.

Installation: Prerequisites

1. Generate GitHub PAT

A Personal Access Token is required for the tool to mine GitHub.

- 1 Login to GitHub > **Settings** > **Developer settings**.
2. Select **Personal access tokens (classic)**.
3. Click **Generate token**.
- 4 Select "No expiration" (optional).
- 5 **Permissions:** repo, write:discuss, project, user.
- 6 **Copy and store it safely!**



Installation: Docker Images

For ARM64 (Apple Silicon)

```
docker pull leopoldotodisco/guido-backend:latest  
docker pull leopoldotodisco/guido-frontend:latest
```

Use these images if you are on an M1/M2/M3 Mac.

For x86 (Intel/AMD)

```
docker pull benedettoscala/guido-backend  
docker pull benedettoscala/guido-frontend
```

Use these images for standard Windows/Linux PCs.

Configuration & Launch

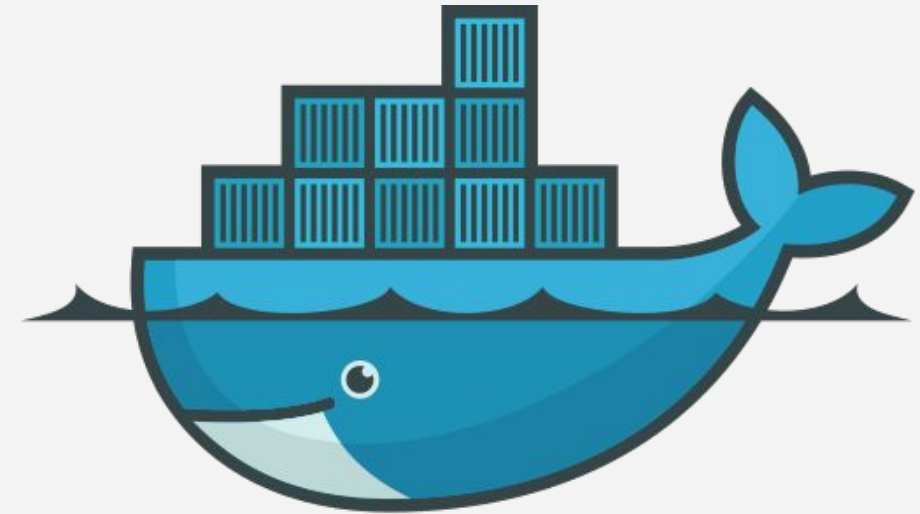
Backend Setup

- 1 Open Docker Desktop > **Images**.
2. Click **Play** on the backend image.
3. **Env Variables:** Add `PAT` with your token value.
- 4 **Port:** Set to `5005`.

Frontend Setup

- 1 Click **Play** on the frontend image.
2. **Port:** Set to `3000`.

If you prefer to build from source or modify the code:



docker

```
# 1. Clone the repository  
# 2. Go to "docker" folder  
# 3. Add PAT in Dockerfile
```

```
docker-compose build  
docker compose up
```

Using GUiDO: Diagnosis Workflow

1

Launch

Open localhost:3000 in browser.

2

Input

Paste GitHub URL into Chat interface.

3

Analysis

System detects smells (I1) & suggests
fixes (I3).

4

Dashboard

Switch view to check cultural
dispersion (I4).

Questions?



Thank you for your attention.

Code: **github.com/StefanoLambiase/GUIDO**